

What Does the Contrary-Hint Score Measure?

A Reproduction of FAITHLM and a Vietnamese Stress Test

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Abstract

FAITHLM evaluates whether a natural-language explanation is faithful by negating it and measuring how much the target model’s prediction shifts. We reproduce its COPA setting with the paper’s own models and obtain a fidelity of 0.872 against the ~ 0.85 reported, then ask what that number measures. Two independently designed controls, absent from the original work, give the same answer: a hint that is *irrelevant* to the question already flips the target on 55.4% of instances, and a hint written for a *different* question flips it on 14.0%, so a large share of the reported score reflects how readily the target follows any hint rather than what the explanation contains. We further show that the released implementation reads predictions by regular-expression matching over generated text, discarding 45–47% of Vietnamese predictions as unparseable. Replacing that step with the probability reading the paper itself specifies raises the flip rate from 0.374 to 0.509 ($p = 0.00011$), and a control isolates the parser, not the continuous signal, as the source of the gain. On Vietnamese XCOPA the choice of target model moves accuracy from 11.8% to 92.5%, which dominates every effect the metric reports.

1 Introduction

Explanations produced by large language models are fluent but not necessarily *faithful*: they need not describe the computation that produced the answer (Jacovi and Goldberg, 2020; Turpin et al., 2023). FAITHLM (Chuang et al., 2026) proposes to measure faithfulness by intervention. Given an explanation E_{NL} of a prediction, an explainer LLM writes a *contrary hint* $\neg E_{NL}$ asserting the opposite, and fidelity is the resulting shift in the target model’s output,

$$S_E := f(X) - f(X \mid \neg E_{NL}). \quad (1)$$

A large shift is read as evidence that the explanation carried decision-relevant content.

We reproduce this measurement and then interrogate it. Our contributions:

1. **Reproduction.** Using the paper’s own target and explainer, the COPA fidelity we obtain matches the published curve (§6.1).
2. **The score is not specific to the explanation.** Two controls designed independently by different authors of this report—an off-topic hint and a hint written for another question—agree that a substantial fraction of the score survives when the explanation’s content is removed (§6.2).
3. **The reader, not the metric, breaks under language transfer.** The released code resolves predictions with a regular expression over generated text. Reading probabilities instead—as §3.1 of the original specifies—recovers a significantly larger effect (§6.3).
4. **A per-language XCOPA evaluation.** The original transfers COPA to XCOPA without naming a language; we report Vietnamese explicitly and show the target model dominates (§6.4).

2 Related Work

Post-hoc explanation of black-box LLMs is usually evaluated either by *simulatability*—can a model reproduce predictions from the explanation (Madsen et al., 2024)—or by perturbing the input and observing the output. Jacovi and Goldberg (2020) argue these test different properties and should not be conflated. Chain-of-thought text is a tempting proxy for reasoning but is demonstrably unfaithful in controlled settings (Turpin et al., 2023; Lanham et al., 2023). FAITHLM sidesteps token-level deletion, which is ill-defined when the decision-relevant content is latent rather than

present in the input, by intervening at the level of meaning. Our concern is not with that move but with the absence of a null condition against which the resulting shift is compared.

3 Dataset

Balanced COPA. Roemmele et al. (2011) pose binary causal commonsense questions; Kavumba et al. (2019) add mirrored instances to suppress lexical shortcuts. We use the 500-item test split.

XCOPA-vi. Ponti et al. (2020) translate COPA into 11 languages with professional translators. Crucially, the Vietnamese test split contains *the same 500 items* as the English split, verified item-by-item, so the two form a parallel corpus in which language is the only variable. This makes the cross-lingual comparison controlled rather than merely suggestive. Neither dataset carries gold explanations, which is why we cannot report the original’s second metric (§8).

4 Method

The measurement. For each instance the target answers the question; an explainer proposes E_{NL} from the (question, answer) pair alone; the explainer then writes $\neg E_{NL}$; the target answers again with $\neg E_{NL}$ prepended as a hint. Fidelity is Eq. 1. An optimisation loop rewrites the explanation to maximise this score, terminating early once a non-zero score appears.

Control A: off-topic hint. We score a third prompt, identical in shape, whose hint is unrelated to the question (“The weather forecast mentions scattered clouds tomorrow afternoon.”). Any flip here cannot come from the explanation’s content.

Control B: mismatched hint. A stricter variant injects the contrary hint written for a *different* instance. It matches the register, length and form of a real hint and differs only in which question it belongs to, so it cannot be dismissed as trivially ignorable.

Score modes. The released code computes $|\Delta\text{accuracy}|$ over an argmax answer parsed out of generated text. We add two alternatives that keep Eq. 1 but change how f is read: PROB_ACCURACY, which takes the argmax over choice probabilities, and LOGPROB, which uses the continuous probability shift.

	COPA-en	XCOPA-vi
Fidelity	0.860	0.336
Control A (off-topic)	0.554	0.122
Corrected	+0.306	+0.214

Table 1: Control A, 500 instances each, Phi-2 target. Roughly 64% of the English score survives when the hint has nothing to do with the question.

5 Experimental Setup

We follow Table 2 of the original for COPA: 20 optimisation steps, target temperature 0.7, explainer temperature 0.9, top- p 0.9, 500 instances. Target microsoft/phi-2 and explainer gpt-3.5-turbo are the paper’s own choices (Javaheripi and Bubeck, 2023). Because both of the original’s explainers have since been retired, a separate axis substitutes four current explainers with everything else fixed. Additional targets (Qwen3.5-4B, DeepSeek-v4-flash) probe the Vietnamese setting. All runs, raw per-instance outputs and the code are released with this report.

Two implementation details of the released code affect any reproduction. First, do_sample is never set, so transformers silently ignores the temperature the paper specifies and decodes greedily; we verified that both regimes give the same result, but a reproduction that follows Table 2 is not running the released configuration. Second, the early-stopping condition is only tested every fifth iteration, so 16.6% of executed iterations occur after the stopping criterion is already satisfied.

6 Results and Discussion

6.1 The reproduction succeeds

On Balanced COPA with Phi-2 and GPT-3.5, fidelity reaches 0.872 over 500 instances against the ~ 0.85 the original reports for COPA. Substituting a modern explainer under the same protocol gives 0.860. The measurement is reproducible.

6.2 But the score is largely non-specific

The contrary hint flipped *every* parseable prediction: all 354 correct answers and all 34 incorrect ones score exactly 1.0. A metric at its ceiling cannot rank explanations, and saturation is equally consistent with a target that simply follows hints.

Table 1 shows Control A. Control B, run independently on a different target and explainer, agrees: an own-question hint flips 0.370 of instances against 0.140 for a hint belonging to an-

Score mode	Flip rate	95% CI
ACCURACY (released)	0.374	[0.315, 0.432]
PROB_ACCURACY	0.460	[0.400, 0.520]
LOGPROB	0.509	[0.449, 0.570]

Table 2: Qwen3.5-4B + GPT-4o-mini, XCOPA-vi, 500 instances. McNemar against the released mode: $p = 0.00011$ (LOGPROB), $p = 0.01003$ (PROB_ACCURACY).

Target	Prompt	Acc.	Unparsed
Phi-2	VI	11.8%	45.4%
Phi-2	EN	41.5%	41.5%
Qwen3.5-4B	EN	90.5%	6.5%
DeepSeek-v4-flash	EN	92.5%	2.0%

Table 3: XCOPA-vi accuracy by target and prompt-scaffold language.

other question ($p < 0.0001$, $n = 200$). The two controls differ in design, target and explainer yet land in the same place—baselines of 0.122 and 0.140, corrected effects of +0.214 and +0.230. The effect is real and semantic, but roughly one flip in seven is noise, and on English roughly two thirds of the headline number is suggestibility.

6.3 The text reader is the dominant confound

The released implementation ends its prompt at ### Response: without stating an output format, then searches the continuation for [choice]. . . @. On XCOPA-vi this discards 45–47% of predictions as unparsable, at the same rate for Phi-2 and Qwen3.5-4B—a prompt–parser mismatch, not a target failure. Where a prediction does survive, the parser disagrees with a normalised argmax on 59% of 150 audited instances.

Table 2 shows what changes when f is read from probabilities. The gain is significant, but the informative comparison is PROB_ACCURACY against LOGPROB: $p = 0.103$ at $n = 500$, and since 500 is the entire test split this is a ceiling rather than a sample-size limitation. *Most of the improvement comes from abandoning the text parser, not from the continuous signal.* Reporting the headline gain without this decomposition would attribute it to the wrong change.

6.4 On Vietnamese, the target model dominates

Table 3 makes the ordering plain: the target model moves accuracy by 80 points, far more than any effect the fidelity metric reports. Phi-2 at 11.8%

on a two-choice task is *below chance* (95% CI [9.3, 14.9]; restricted to the 273 parseable answers, 21.6%, CI [17.1, 26.9]), so it is selecting the wrong option systematically rather than guessing. Control A collapses in step (0.554 $\rightarrow$ 0.122), which is coherent: a model that cannot read the question cannot read the hint either.

Splitting by whether the target answered at all reverses the aggregate picture. Among instances with a correct prediction the corrected effect is *larger* in Vietnamese (+0.475, $n = 59$) than in English (+0.401, $n = 354$); the unparsable instances contribute +0.044 and drag the mean down. The fidelity measure survives the language change; the reader does not.

Swapping the explainer across four laboratories with everything else fixed moves fidelity only between 0.930 and 0.975, so the metric is far more sensitive to how predictions are read than to who writes the explanation.

6.5 Cost scales inversely with success

Vietnamese required 7.82 optimisation iterations per instance against 3.11 for English. Early stopping fires on the first non-zero score, so when the measurement yields nothing every instance runs to the cap. The framework spends the most compute exactly where it works least—a sharper form of the computational cost the original raises in its own limitations.

7 Error Analysis

Of 14 errors across both English runs, 13 are parsing failures and one is a genuinely wrong choice. What the pipeline calls accuracy is closer to a parse-success rate.

The failure cascades. When the target’s answer is unresolvable the pipeline passes the literal string X to the explainer, which duly explains it: “*The model chose answer X based on the content of the question. . .*” The explanation, the contrary hint derived from it, and the resulting score are then all about an answer that was never produced. This is why unparsable instances show a corrected effect of only +0.018—+0.044: there is nothing for an explanation to be faithful to, and the control correctly detects it.

Inspecting raw generations shows Phi-2 answering the plain prompt correctly in English but, once Let’s think step by step. is appended for scoring, producing textbook-style exposition

(“*Use Case 1: Shipping Fragile Items*”) that never states a choice. In Vietnamese it degenerates further, emitting corrupted tokens (“*Lê nhiều*”, “*Bnh hi*”) in place of the scaffold words. Both behaviours are invisible to a metric that reads only a parsed answer.

8 Limitations and Ethical Considerations

Single repetition. The original averages three runs; ours are single runs, so we report no variance. This is the same gap we identify in the original, and we flag it rather than repeat it silently.

Missing second metric and baselines. We do not report truthfulness, which requires gold explanations that COPA and XCOPA lack; ECQA (Aggarwal et al., 2021) would be needed. The original’s baselines (Madsen et al., 2024; Wang et al., 2022) are absent from the released code and we did not reimplement them, so our numbers compare against the original’s published values rather than against re-run baselines.

Granularity. Under the released mode S_E carries one bit per instance, since accuracy over a single example is 0 or 1. Small samples are uninformative: a 30-instance estimate of Vietnamese accuracy gave 33.3% with an interval that still contained chance, against 11.8% at $n = 500$. We twice drew conclusions from $n = 2$ that the full runs overturned.

Ethics. All datasets are public research corpora. Explainer calls were served through an institutional gateway; a small number of instances containing violent lexical items returned empty completions from content filtering, which we count and disclose rather than silently score as wrong. No human subjects were involved.

9 Conclusion

FAITHLM reproduces. Its contrary-hint score, however, mixes two things that the original does not separate: whether the explanation carried the target’s reason, and whether the target follows hints at all. Two independently designed controls agree that the second accounts for a large share of the reported number. Under language transfer the binding constraint is not the metric but the regular-expression reader in front of it, and reading probabilities—as the original’s own formulation specifies—recovers a significantly larger effect. We recommend that intervention-based faith-

fulness measures report a null-hint baseline alongside the headline score, and that the quantity being intervened on be read from the model’s distribution rather than from its prose.

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